

CS1CA Computer Systems Architecture  
*Storage systems and networking (1)*  
Week 12 lecture 1 of autumn semester 2025

*Lectured by*

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This topic is delivered in week 12  
of autumn semester 2025

# Outline of contents in the topic

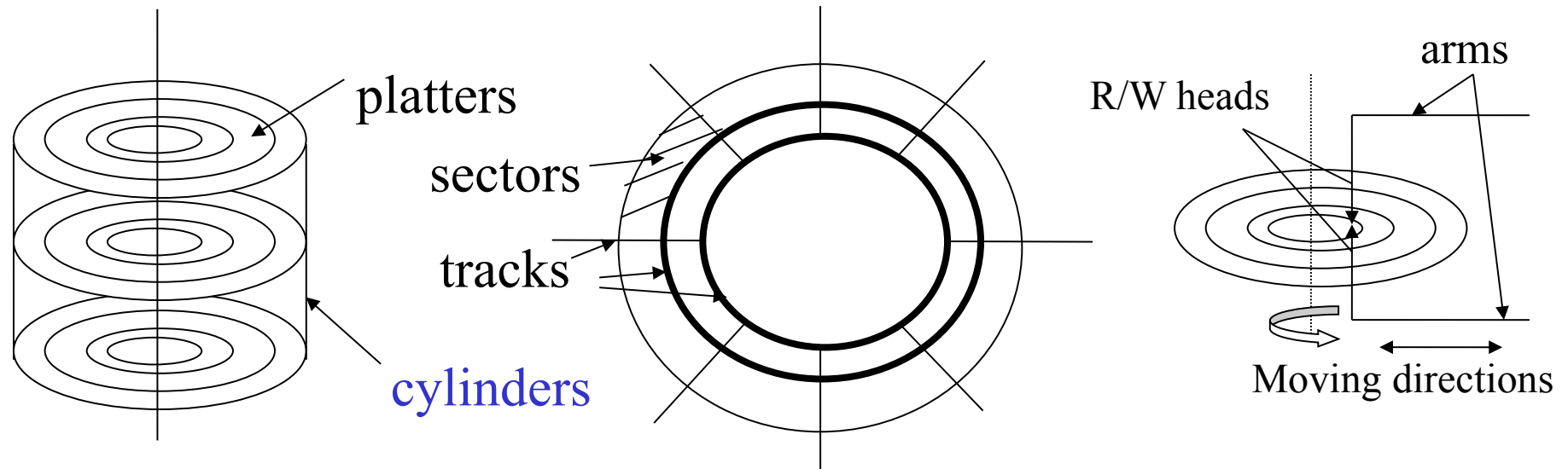
- Magnetic disk (Solid State Drive, SSD, is not covered in the topic, but you are encouraged to learn it by yourself)
  - Structure and management
  - Disk performance measure
- Buses
- I/O and I/O performance measures
- Reliability, availability, and RAID
- A 2-node simple network: performance of the simple network

# In this lecture

- The following contents will be delivered
  - Magnetic disk (Solid State Drive, SSD, is not covered in the topic, but you are encouraged to learn it by yourself)
    - Structure and management
    - Disk performance measure
  - Buses
- Expected Learning outcomes are listed below.
  - To gain knowledge of structure and organization of magnetic disks, and to be able to measure disk performance
  - To have the knowledge of bus design options

# Disks: magnetic disks (1)

- Magnetic disks work as secondary storage devices for long term **non-volatile storage** for files.
- Disks are organised into **platters, cylinders, tracks, and sectors**.

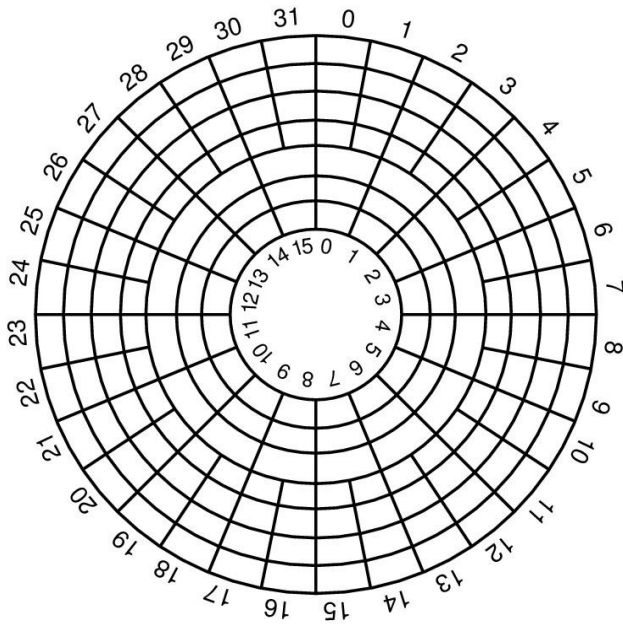


## Disks: magnetic disks (2)

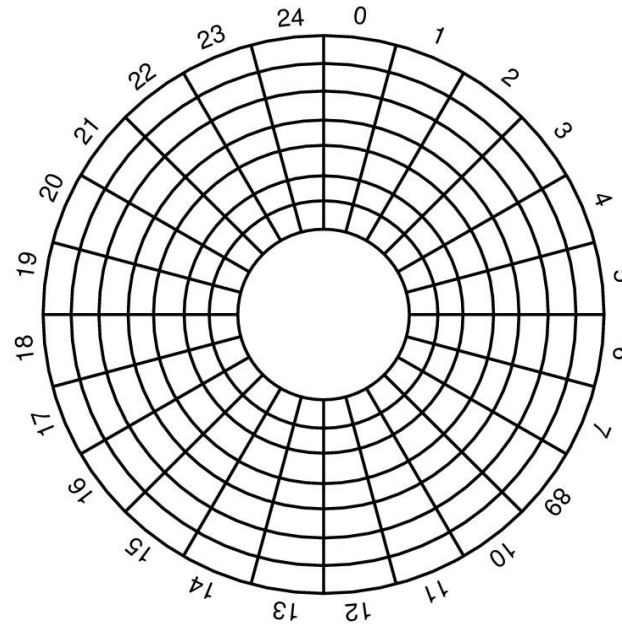
- Each cylinder contains a number of tracks ranging from 1 to 16 depending on the number of R/W heads stacked vertically.
- A track is divided into sectors, with the number of sectors along the circumferences.
- On older disks, all tracks have the same number of sectors. The outer tracks, which are longer, have a lower density than the inner tracks.
- Modern disks use the concept of track zones with more sectors on the outer zones than the inner ones.

# Disks: magnetic disks (3)

- Physical geometry of a disk with two zones
- A possible virtual geometry for this disk.



Physical geometry



Virtual geometry

# Disks: magnetic disks (4)

- A virtual disk geometry is presented to the OS to hide the different number of sectors for each track.
- Disk drives are addressed as a large one-dimension array of logical blocks.
- Map the logic block array onto the sectors of the disk
  - Sector 0 is the first sector of the first track, may be on the outermost cylinder.
  - The mapping proceeds in order through that track, then through the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost.

# Magnetic disk performance measure

- *Seek time*: the time to move arm to desired track;
- *Rotational delay*: the time for requested sector to rotate under the head (rpm – revolution per minute);
- *Transfer time*: the time to transfer a block of bits (a sector typically) under the head;
- *Controller overhead*: the time the controller imposes in performing an I/O access;
- *Queuing time*: the time spent waiting for a disk to be available.

$$\text{Average disk access time} = \text{seek time} + \text{rotational delay} + \text{transfer time} + \text{controller overhead} + \text{queuing time}$$

## Example

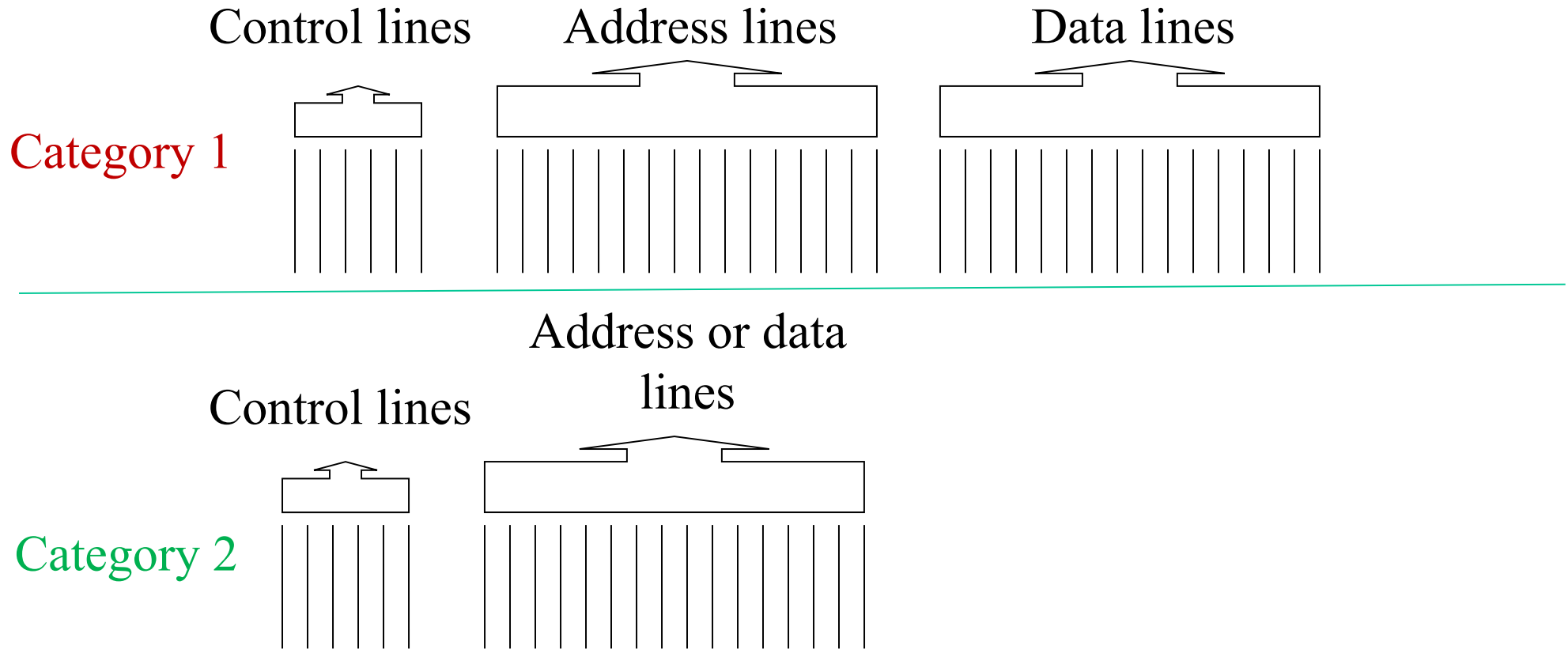
- Given information
  - A 512 byte sector; seek time = 9 ms (ms: millisecond)
  - Transfer rate = 4 MB/s
  - rotation speed = 7200 rpm (revolution per minute)
  - Controller overhead = 1 ms; no queuing delay
- Question: what is the average disk access time?
- A.:

*Average disk access time*

# Buses: connecting I/O devices to CPU/memory

- Bus serves as a shared communication link between subsystems;
- Two types of buses
  - I/O buses: lengthy; cope with a wide range in the data bandwidth; follow a bus standard.
  - CPU/memory buses: short; high speed; match to the memory system to maximise memory-CPU bandwidth.

# Buses: address, control and data lines



- Two categories: **separate** and **shared** address and data lines

# Bus transactions

- A bus transaction includes two parts: sending the address and receiving or sending the data.
- Two definitions to bus transactions depending on what they do to memory.
  - A read transaction transfers data from memory to either the CPU or an I/O device; send address, send control signal indicating read, memory returning data on bus, CPU wait between sending address and receiving data.
  - A write transaction writes data to the memory; send both address and data, no waiting.

# Bus design options

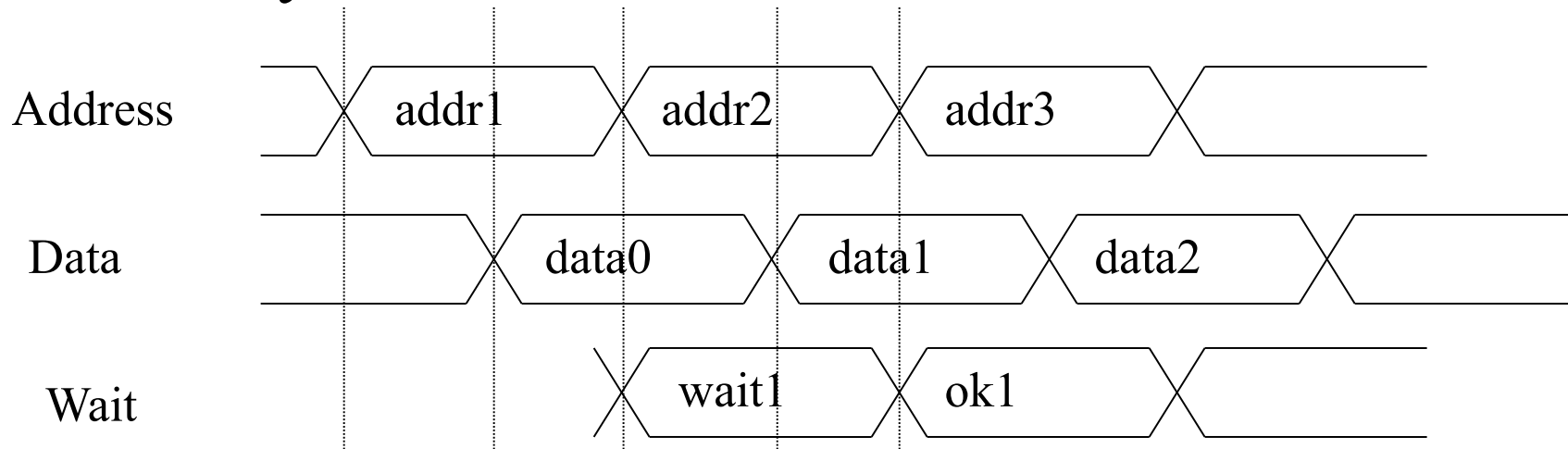
| Bus options         | High performance                              | Low cost                          |
|---------------------|-----------------------------------------------|-----------------------------------|
| Bus category        | Separate address and data lines               | Shared address and data lines     |
| Data width          | Wider is faster (e.g. 64 bits)                | Narrower is cheaper (e.g. 8 bits) |
| Transfer size       | Multiple words: less bus overhead             | Single word transfer is simpler   |
| Bus master          | Multiple masters (arbitration schemes needed) | Single master                     |
| Split transactions? | Yes: packets                                  | No: continuous connection         |
| Clocking            | Synchronous                                   | Asynchronous                      |

# Bus masters

- Bus masters are devices that can initiate a read or write transaction.
- CPU is always a bus master.
- Multiple bus masters: multiple CPUs or I/O devices can initiate a bus transaction.
- Arbitration scheme used for multiple bus masters to decide who gets the bus next.
  - Priority scheme
  - Randomly selecting scheme

# Split transactions

- Split transaction bus can provide higher bandwidth by using packets rather than holding the bus for the full transaction;
- Each transaction must be tagged so that CPU and memory can tell what is what.



# Clocking: synchronous/asynchronous bus

- Synchronous bus:
  - Central clock - therefore same clock rate on the control lines;
  - Simple control logic;
  - Fast – a fixed protocol;
  - Skewing problems (clock skew accumulates in a long bus).
- Asynchronous bus:
  - Self-timed: each component may have its own clock;
  - Hand-shaking protocols;
  - Accommodates a wide variety of devices;
  - No clock skew or synchronisation problems;
  - Slow.

# Bus standards

- Bus standards make it possible that computer designer and I/O device designer work independently.
- **Bottom-up** standards: from manufacturers to a sanctioning body like **ANSI** or **IEEE**;
- **Top-down**: from a sanctioning body to manufacturers, such as PCI-bus standards.
- For SCSI drive, please refer to the website of <https://www.computerhope.com/jargon/s/scsi.htm> for details.

# End of this lecture

## Questions and exercises

- What is the basic structure (organization) of a magnetic disk? How to measure its performance?
- What are bus design options?